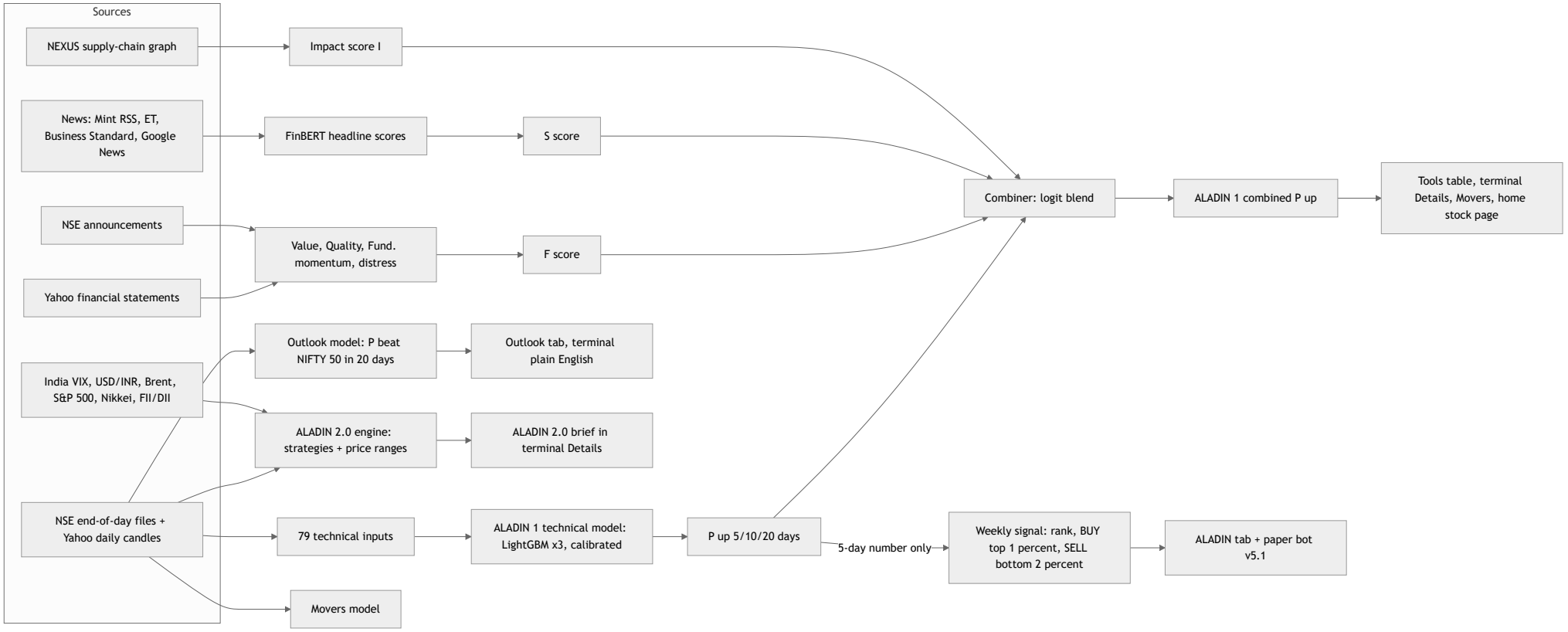
How ALADIN works today

Written 2026-11-08 from the code on `main` and the data published for the 2026-11-06 close. Nothing was changed. Words in *italics* are explained in GLOSSARY.md.

1.1 The big picture

ALADIN is five prediction systems that share data. Only **ALADIN 1** blends the three fronts you know (technical, fundamental, sentiment). The **ALADIN tab** shows a different system (the weekly signal), which uses ALADIN 1's technical 5-day number and nothing else.



Walkthrough. Every evening after the market closes the *Daily update* job saves the day's prices and runs the Outlook model. When it ends it starts *ALADIN nightly*: it refreshes about 400 companies' statements, scores filings, computes the supply-chain impact, builds 79 technical inputs for every stock, runs the technical model

for 5, 10 and 20 days, converts its raw score into a *calibrated* probability, and blends it with the fundamental and sentiment scores. Then *ALADIN2 nightly* starts: it runs the ALADIN 2.0 engine, writes the weekly BUY/SELL list from the technical 5-day number, and updates the paper bot. The sentiment job runs separately about every 30 minutes. The website reads the saved files; the colour you see is chosen in the browser by fixed cut-offs (section 1.5).

1.2 Data, source by source

Source	Gives	Fetched by	Updates	Usual age when used	When missing or late
NSE end-of-day files + Yahoo candles	daily open/high/low/close/volume, split-adjusted (not dividend-adjusted)	nse_eod.py, terminal_hist.py (Daily update)	every evening	same day's close	a stock whose last bar is >10 days old is not scored (aladin_model.py)
Yahoo statements	annual + quarterly income, balance sheet	aladin_fund.py, ~400 companies a night	rotating; each company every few nights	days to weeks; **filing dates not stored**	F is None, its weight drops out of the blend (counted as 0). RELIANCE has no F today
NSE announcements	text of filings, scored by FinBERT	aladin_filings.py	nightly, last 90 days, 15-minute budget	1 day	the NLP part of F is left out
News	headlines	aladin_sentiment_engine.py, news_wire.py	about every 30 min	minutes to hours; decays with a 24-hour half-life-like weight	S is None, counted as 0 in the blend (most stocks: only 2 headlines for RELIANCE, none for PERSISTENT, SAILIFE)
Market series (VIX, INR, Brent, S&P 500, Nikkei, US 10y)	context for ALADIN 2.0	ALADIN2 nightly	daily	0–1 day	the feature is empty that day
FII/DII flows	net buying	institutional.py	daily	1 day	last saved day kept; used inside sentiment as "flow"
NEXUS graph	supply-chain links	nexus.yml, graph_impact.py	weekly	days	I is None (true for almost every stock)
Liquidity sweeps	live price patterns	the owner's PC only	live	–	absent on GitHub, so always 0 in the nightly blend

Silent fills. In the blend, a missing F, S, I or sweep counts as **0** (no push either way). The technical model itself lets LightGBM handle missing inputs (it sends them down a learned branch). Nothing is filled with an old value except prices (last saved copy).

1.3 Inputs ("features")

Technical, 79 inputs in total (scripts/predict_model.py `features`, scripts/aladin_model.py `stock_frame`, `build_dataset`):

Group	Inputs (code name: meaning)	Look-back	Scaling
Returns	r1, r5, r10, r20, r60, r120, r250: log change in price over 1 day to 1 year; mom12_1: 1-year minus 1-month return	1–250 days	raw; several also ranked 0–1 across stocks that day (`_rk`)
Trend	d_sma20/50/200: price ÷ its 20/50/200-day average − 1; sma50_slope, sma200_slope: change of those averages over 10/20 days	20–220 days	raw
Oscillators	rsi2, rsi14 (relative strength 0–100), stoch14 (where the close sits in the 14-day range), cci_z, adx (trend strength), di_diff, macd_h, bb_pct, bb_w	2–26 days	raw
Volatility	atr_pct (typical daily range ÷ price), vol20, vol60, vol_ratio, skew60, range20, gap20	14–60 days	raw
Position	hi52, lo52 (distance from 52-week high/low), dd60 (fall from the 60-day high)	60–252 days	raw
Volume	vz20 (today's volume vs its 20-day average, log), vtrend, turn (traded value)	5–60 days	raw
Market link	beta, corr60, idio60 (stock-specific volatility), rel20, rel60 (return vs NIFTY 50), sec_rel60 (vs its sector)	20–250 days	raw
Market-wide	m_r5/20/60 (NIFTY returns), m_d200, m_vol20, m_br50/m_br200 (share of stocks above their 50/200-day average), m_disp	5–200 days	same for every stock that day
Quant extras	pca_z, pca_r2 (move not explained by market factors), coint_z, coint_hl, coint_p (gap to a paired stock), hmm_p_highvol (chance the market is in a turbulent state), jump_lam, jump_recent, jump_bv	20–250 days	raw
Ranks	17 of the above ranked 0 (lowest) to 1 (highest) among that day's liquid stocks	–	percentile
Labels	ind: industry (category)	–	category
Calendar	**moy: month of the year (1–12); dom: day of the month (1–31)**	–	raw. Added because it "improved 8 of 13 test years"

Fundamental (aladin_fund.py): value, quality, fundamental momentum (each a z-score, compared across companies), NLP tone of filings, distress band. `F = 100 × tanh(z ÷ 1.5)` where `z = 0.35 value + 0.35 quality + 0.20 momentum

- 0.10 NLP − distress penalty` (weights re-scaled over the parts present). Range −100 to +100.

Sentiment (aladin_sentiment_engine.py): each headline gets FinBERT's P(positive) − P(negative); weighted by age (`exp(-hours ÷ 24)`) and source; per stock `raw = 0.60 news + 0.15 retail + 0.15 FII/DII flow` (re-scaled over the parts present) **plus a hand-set geopolitical adjustment** (`geo_adjust` , from geo_exposure.json); `S = 100 × raw` , clipped to ±100. Levels: BAD ≤ −60, POOR ≤ −20, NEUTRAL, GOOD ≥ 20, EXCELLENT ≥ 60.

1.4 Each model

	ALADIN 1 technical	Outlook	ALADIN 2.0 engine	Weekly signal	Movers
Type	machine learning: 3 LightGBM tree models (different random seeds), averaged; for 10 days a calm/stressed blend	machine learning: 3 LightGBM classifiers + 1 regressor, rank-averaged	strategy library (trend, reversal, volatility, relative strength, delivery), strategy life cycle, volatility model for ranges	a rule on ALADIN 1's number	machine learning
Predicts ("target")	close in 5/10/20 trading days > today's close (raw direction, **no costs**, not vs the market)	20-day return beats NIFTY 50's	per strategy: net return after costs; ranges: price quantiles	next 5 days, enter next open	5-day move size and direction
Trained on	2008/2009 onward, main-board stocks with ≥ ₹5 Cr a day **on that day** (point in time); walk-forward, test years 2013–2026 with a 15-day gap; final model retrained nightly on everything	NIFTY 500 **as of today** (today's list), 2008 onward, walk-forward 2013–2026, 25-day gap	2005 onward, test from 2012-07, 63-day blocks, 20-day purge + 5-day embargo	the weekly study 2013–2026 (310,816 stock-weeks)	2016–2026
Outputs	calibrated probability 2–98%	calibrated probability, decile	product state; forecast range; signal only if Validated	BUY / SELL / HOLD, a stop, a "target"	size + direction
Thresholds and where from	calibration: isotonic map fitted on the 2013–2026 unseen-year record; confidence Low < 3 pts from 50%, Medium < 7, else High	none for the colour	Validated: ≥ 60 unseen trades, ≥ 60 live days, calibration error ≤ 5%, net ≥ 5 bps; kill switch at calibration error > 5%	BUY ≥ 99th percentile, SELL ≤ 2nd (tested: chosen on 2013–2020, held on 2021–2026); stop = 2 × ATR	pre-declared gates
Weight in a blend	the base of the ALADIN 1 blend	–	–	–	–
Measured skill (unseen data)	AUC 0.546 at 10 days (0.5 = chance); top tenth went up 54.3%, bottom tenth 43.5%	AUC 0.534	live: **suspended** (see 1.7)	BUY top 1%: +27 bps a week after costs in the study	direction AUC 0.511: **failed**

The ALADIN 1 blend (scripts/aladin_model.py `combine_py` , identical in desk-aladin.js): `logit(p) = logit(p_technical) + 0.20 × F_adj/100 + 0.12 × S/100 + 0.18 × sweep/100` , with `F_adj = F − I` . Weights 0.20 / 0.12 / 0.18 are **set by hand ("prior")**: a fitted blend is only tried after 120 trading days of matured live predictions (16

so far). Missing parts count as 0. Result clipped to 2–98%.

1.5 From a number to a colour

Page	Number shown	Green	Grey	Red
ALADIN tab, signal badge	weekly signal	BUY (rank ≥ 99th percentile)	HOLD	SELL (rank ≤ 2nd)
ALADIN tab, "stop loss" cell	distance of the stop from the price	stop above price	–	**stop below price: always, for every BUY**
ALADIN tab, "target" cell	distance of the range edge	above price	–	below price ("not measured" for 12 of 13 BUYs this week)
Terminal Details, Tools table (`probBar`), Movers	ALADIN 1 combined P(up), 10 days	≥ 55%	45–55%	≤ 45%
Home stock page, "What our models see"	ALADIN 1 technical P(up) at 5, 10, 20 days	**≥ 50%**	–	**< 50%**
Outlook tab, terminal	P(beat NIFTY 50)	≥ 55%	45–55%	≤ 45%
ALADIN 2.0 brief	product state	Validated + signal	Provisional / HOLD	–

Today's spread of ALADIN 1 (2026-11-06 close, 2,458 stocks).

Horizon	Lowest	Median	Highest	Stocks above 50%	Different values
5 days (technical)	45.3%	51.6%	52.4%	85.6%	17
10 days (technical)	41.0%	**44.5%**	**50.0%**	**0%**	21
20 days (technical)	45.9%	48.5%	53.8%	16.3%	19
10 days (combined)	–	44.5%	–	0.3%	–

What this means: on every page that colours ALADIN 1's 10-day number, no stock can be green, and on the home stock page every stock's 10-day bar is red. The 13 BUY stocks on the ALADIN tab all have the same 5-day value, 52.4%.

1.6 Three worked examples (2026-11-06 close)

RELIANCE (large-cap)

Step	Numbers	What happened
Raw data	close ₹1,320; +6.8% vs 50-day average, −0.4% vs 200-day; 1 month +9.3%, 1 year −11.3%; 82% of its 52-week high; volume 1.09× normal	a recovering large-cap, flat on the year
Technical model	P(up) 5d 51.6%, **10d 44.3%**, 20d 49.5%	the 10-day number is low
Biggest reasons (10-day, in log-odds)	**day of month −0.245**, industry +0.042, month of year +0.039, 1-week return rank −0.024, RSI(2) −0.020	the calendar alone outweighs everything about the company
Fundamental	F = **none** (no statements in the data)	counted as 0
Sentiment	S = +28.9 (GOOD, 2 headlines)	small push up
Blend, 10 days	logit(0.443) = −0.229; + 0.12 × 28.9/100 = +0.035; total −0.194 → **45.2%** (published 45.05%)	blend reproduced by hand
Colour	Terminal: grey (45–55%). Home stock page: 10-day bar **red**. Weekly rank 50th percentile: HOLD	

PERSISTENT (mid-cap)

Step	Numbers	What happened
Raw data	close ₹6,053.5; +7.1% vs 50-day, +13.8% vs 200-day; 1 month +9.9%, 3 months +10.9%; 92% of its 52-week high	in an uptrend
Technical model	5d 52.4%, **10d 46.8%**, 20d 52.7%	
Biggest reasons	**day of month −0.240**, industry +0.139, month +0.030, 1-week return rank +0.019, 52-week-high rank +0.013	calendar again the largest
Fundamental	value −0.87, quality +0.06, momentum +0.17, distress Low → F = −21.8	expensive, so a push down
Sentiment	none	0
Blend, 10 days	−0.128 + 0.20 × (−21.8)/100 = −0.044 → −0.172 → **45.7%** (published 45.7%)	
Colour	Terminal: grey. Home stock page: 10-day bar **red**. Weekly rank 88th percentile: HOLD. Outlook 50.6% (decile 7)	

SAILIFE, Sai Life Sciences (should look bullish)

Step	Numbers	What happened
Raw data	close ₹1,785.1; **98.4% of its 52-week high**; +16.6% vs 50-day, +47.1% vs 200-day; 3 months +26%, 1 year +94%; **volume 11.8× normal**	a textbook breakout on heavy volume
Technical model	5d 51.6%, **10d 44.1%**, 20d 50.6%	reads as below a coin flip
Biggest reasons	**day of month −0.227**, industry +0.101, 1-week return rank −0.026, month +0.026, RSI(2) −0.020	the breakout is not among the top reasons
Fundamental	value −1.07, quality −0.28, momentum +0.99, distress Low → F = −8.2	
Sentiment	none	
Blend, 10 days	−0.237 + 0.20 × (−8.2)/100 = −0.016 → −0.254 → **43.7%** (published 43.65%)	
Colour	Terminal: **red** (≤ 45%). Home stock page: 10-day bar **red**. Weekly rank 39th percentile: HOLD. Outlook 47.6%	

1.7 Health snapshot (2026-11-08)

Item	State
ALADIN 1	probabilities as of 2026-11-06; models retrained every night; unseen-year record and calibration last rebuilt up to 2026-09-15 (10-day)
Outlook	as of 2026-11-06; record to 2026-10-08
ALADIN 2.0 engine	**all 1,382 stocks Suspended by its own kill switch**: live calibration error 14.7% (limit 5%); live 50% range caught the price 44.5% of the time, 80% range 75.2%. So no ALADIN 2.0 signal anywhere
Weekly signal	as of 2026-11-06: 13 BUY, 26 SELL, 1,245 HOLD; all "Provisional" (no live record yet)
Movers	failed its own test (direction AUC 0.511); its live list is not on `main`
Fitted blend	not tried yet: 16 of 120 matured days
Sweep and impact terms	not validated: both effectively 0
Jobs	last 6 runs of ALADIN nightly, ALADIN2 nightly and ALADIN sentiment all succeeded. ALADIN nightly ran 6 times in 15 hours on 7–8 Nov because Daily update ran 5 times (schedule, re-starts and a failed run); each Daily update starts the ALADIN chain. Harmless but wasteful

Item	State
Tests	288 ALADIN 2.0 + ALADIN Python tests pass; 34 of 35 ALADIN 1 model tests pass (the one failure is a timing test that depends on this PC's speed, known before); 33 browser-code tests pass
Data gaps	RELIANCE and others without statements (F none); news covers few stocks; filing dates not stored; past NIFTY 500 lists not available; the Outlook model trains on today's NIFTY 500 list